

## Inference Rules in DBMS

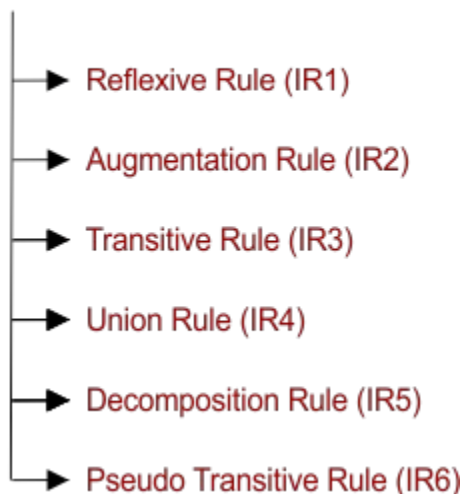
In **Database Management Systems (DBMS)**, **Inference Rules** are used to derive additional **Functional Dependencies (FDs)** from the initial set of attributes. These rules help in simplifying the structure of a database by logically inferring relationships between attributes, ultimately aiding in tasks like **normalization**.

Each rule provides a way to derive new functional dependencies based on existing ones. These rules are vital in **data consistency** and **query optimization** in relational databases.

### Types of Inference Rules

There are six types of **Inference Rules** in DBMS that help in deriving **Functional Dependencies**.

#### **Inference Rule's (IR)**



Let's discuss each one in detail:

Inference rules in databases are also known as Armstrong's Axioms in Functional Dependency.

These rules were introduced by William W. Armstrong.

**Attributes:** When we talk about databases, we think of them as organized collections of information. Imagine that you have a table called "Student." Now, this table has columns, which we also call "Attributes." These columns define specific details about the students. For example:

**Student\_name:** This column stores the names of the students.

**Roll\_no:** Here, we keep track of their roll numbers.

**Marks:** And finally, we record their exam scores.

**Functional Dependencies (FDs)** are like the building blocks of a database.

Imagine you have a bunch of attributes (think of them as characteristics) in a table. These attributes can be related to each other in interesting ways or say logically. For example,  $\text{Roll\_no} \rightarrow \text{Marks}$  means that from Roll\_no we can get

the Marks of the student, which shows that they are Roll\_no is logically related to Marks.

### Inference Rules

There are 6 inference rules, which are defined below:

**Reflexive Rule:** According to this rule, if B is a subset of A then A logically determines B. Formally,  $B \subseteq A$  then  $A \rightarrow B$ .

Example: Let us take an example of the Address (A) of a house, which contains so many parameters like House no, Street no, City etc. These all are the subsets of A. Thus, address (A)  $\rightarrow$  House no. (B).

**Augmentation Rule:** It is also known as Partial dependency. According to this rule, If A logically determines B, then adding any extra attribute doesn't change the basic functional dependency.

Example:  $A \rightarrow B$ , then adding any extra attribute let say C will give  $AC \rightarrow BC$  and doesn't make any change.

**Transitive rule:** Transitive rule states that if A determines B and B determines C, then it can be said that A indirectly determines B.

Example: If  $A \rightarrow B$  and  $B \rightarrow C$  then  $A \rightarrow C$ .

**Union Rule:** Union rule states that If A determines B and C, then A determines BC.

Example: If  $A \rightarrow B$  and  $A \rightarrow C$  then  $A \rightarrow BC$ .

**Decomposition Rule:** It is perfectly reverse of the above Union rule. According to this rule, If A determined BC then it can be decomposed as  $A \rightarrow B$  and  $A \rightarrow C$ .

Example: If  $A \rightarrow BC$  then  $A \rightarrow B$  and  $A \rightarrow C$ .

**Pseudo Transitive Rule:** According to this rule, If A determined B and BC determines D then BC determines D.

Example: If  $A \rightarrow B$  and  $BC \rightarrow D$  then  $AC \rightarrow D$ .

So we can say that Armstrong's Axioms Related to Functional Dependencies

In a relation **R**,

A functional dependency  $X \rightarrow Y$  means:

If two tuples have the same values for X, they must have the same values for Y.

Example: STUDENT(SID, Name, Dept)

If:  $SID \rightarrow Name$  This means:

Each SID determines exactly one Name.

Armstrong's Axioms are **rules used to derive new functional dependencies from given FDs**.

If some FDs are true, what other FDs must also be true?

There are **three fundamental rules**.

Reflexivity (Trivial FD Rule)

Rule: If  $Y \subseteq X$ , then  $X \rightarrow Y$

A set of attributes always determines its subset.

Example: If:  $X = \{SID, Name\}$

Since  $Name \subseteq \{SID, Name\}$

Then:  $\{SID, Name\} \rightarrow Name$

This is called a **trivial functional dependency**.

Every relation automatically satisfies this.

Augmentation Rule: If  $X \rightarrow Y$  Then for any attribute set  $Z$ :

$XZ \rightarrow YZ$

If  $X$  determines  $Y$ , adding the same attributes on both sides keeps it valid.

Example: Given:

$SID \rightarrow Name$  Add Dept to both sides:

$SID, Dept \rightarrow Name, Dept$

Why is this valid?

Because if  $SID$  determines  $Name$ , then even with  $Dept$  added,  $SID$  still determines  $Name$ .

3 Transitivity Rule:

If

$X \rightarrow Y$   $Y \rightarrow Z$  Then  $X \rightarrow Z$

Dependency passes through an intermediate attribute.

Example: Given:  $SID \rightarrow Dept$   $Dept \rightarrow HOD$  Then:  $SID \rightarrow HOD$

This is very important in finding closure.

Derived Rules (From the Above 3)

Using the 3 axioms, we can derive:

Union Rule

If:  $X \rightarrow Y$   $X \rightarrow Z$  Then:  $X \rightarrow YZ$

Example:

$SID \rightarrow Name$   $SID \rightarrow Dept$  Then:  $SID \rightarrow Name, Dept$

Decomposition Rule

If:  $X \rightarrow YZ$  Then:  $X \rightarrow Y$   $X \rightarrow Z$

Pseudotransitivity

If:  $X \rightarrow Y$   $YZ \rightarrow W$  Then:

$XZ \rightarrow W$

Complete FD Example

Relation:

$R(A, B, C, D)$

FDs:

$A \rightarrow B$

$B \rightarrow C$

$C \rightarrow D$

Using Armstrong's Axioms:

$A \rightarrow B$

$B \rightarrow C$

So  $A \rightarrow C$  (Transitivity)

$A \rightarrow C$

$C \rightarrow D$

So  $A \rightarrow D$  (Transitivity)

Now we have:

$A \rightarrow B, C, D$

Therefore:

$A = \{A, B, C, D\}$

So A is a candidate key.

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